

Experiment ‘p_3_4_classification’ Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_3_4_classification

Natural language statement: null

Method signature: p_3_4_classification (i: int, j: int, k: int) returns (s: string)

Ensures

- if i == j == k then s == "all the same" else if i != j && j != k && i != k then s == "all different" else s == "neither"

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_3_4_classification (i: int, j: int, k: int) returns (s: string)`

The method should respect the following contract:

`ensures if i == j == k then s == "all the same" else if i != j && j != k && i != k then s
↪ == "all different" else s == "neither"`

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn39UqXhhBzHnbmHdNUf8iXLv8dxC

Model: gpt-5.5-2026-04-23

Created at: 1780582560

Finish reason: stop

Usage: 332 tokens in, and 200 tokens out

```
{  
  if i == j && j == k {  
    s := "all the same";  
  } else if i != j && j != k && i != k {  
    s := "all different";  
  } else {  
    s := "neither";  
  }  
}
```

Final Program

```
method p_3_4_classification (i: int, j: int, k: int) returns (s: string)  
  ensures if i == j == k then s == "all the same" else if i != j && j != k && i != k  
    ↪ then s == "all different" else s == "neither"  
{  
  if i == j && j == k {  
    s := "all the same";  
  }
```

```
} else if i != j && j != k && i != k {  
  s := "all different";  
} else {  
  s := "neither";  
}  
}
```

Total Token Usage

Input tokens: 332

Output tokens: 200

Reasoning tokens: 116

Sum of 'total tokens': 532

Experiment Timings

Overall Experiment started at 1780582559727, ended at 1780582595512, lasting 35785ms (35.79 seconds)

Iteration #1 started at 1780582559729, ended at 1780582595512, lasting 35783ms (35.78 seconds)